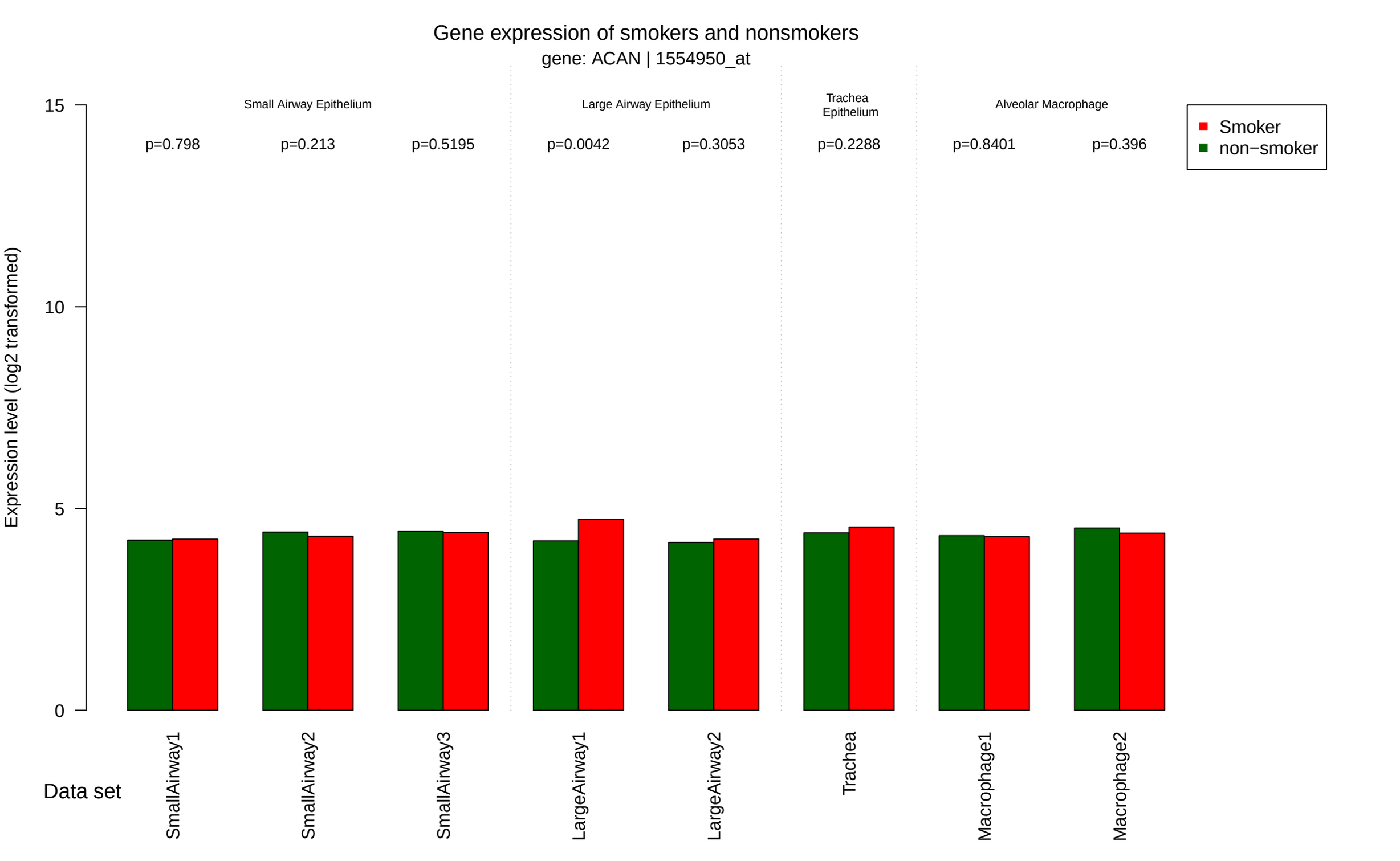
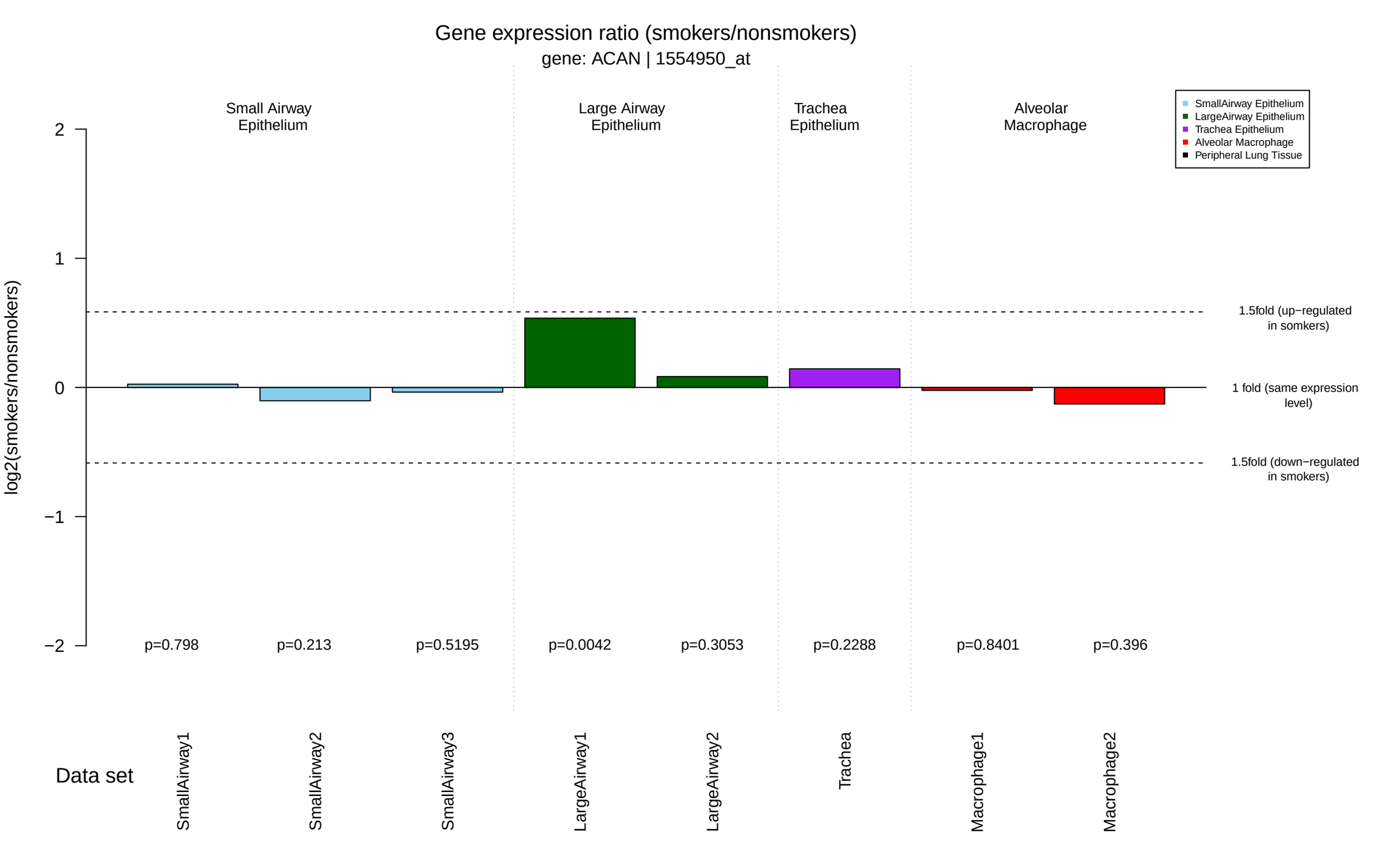
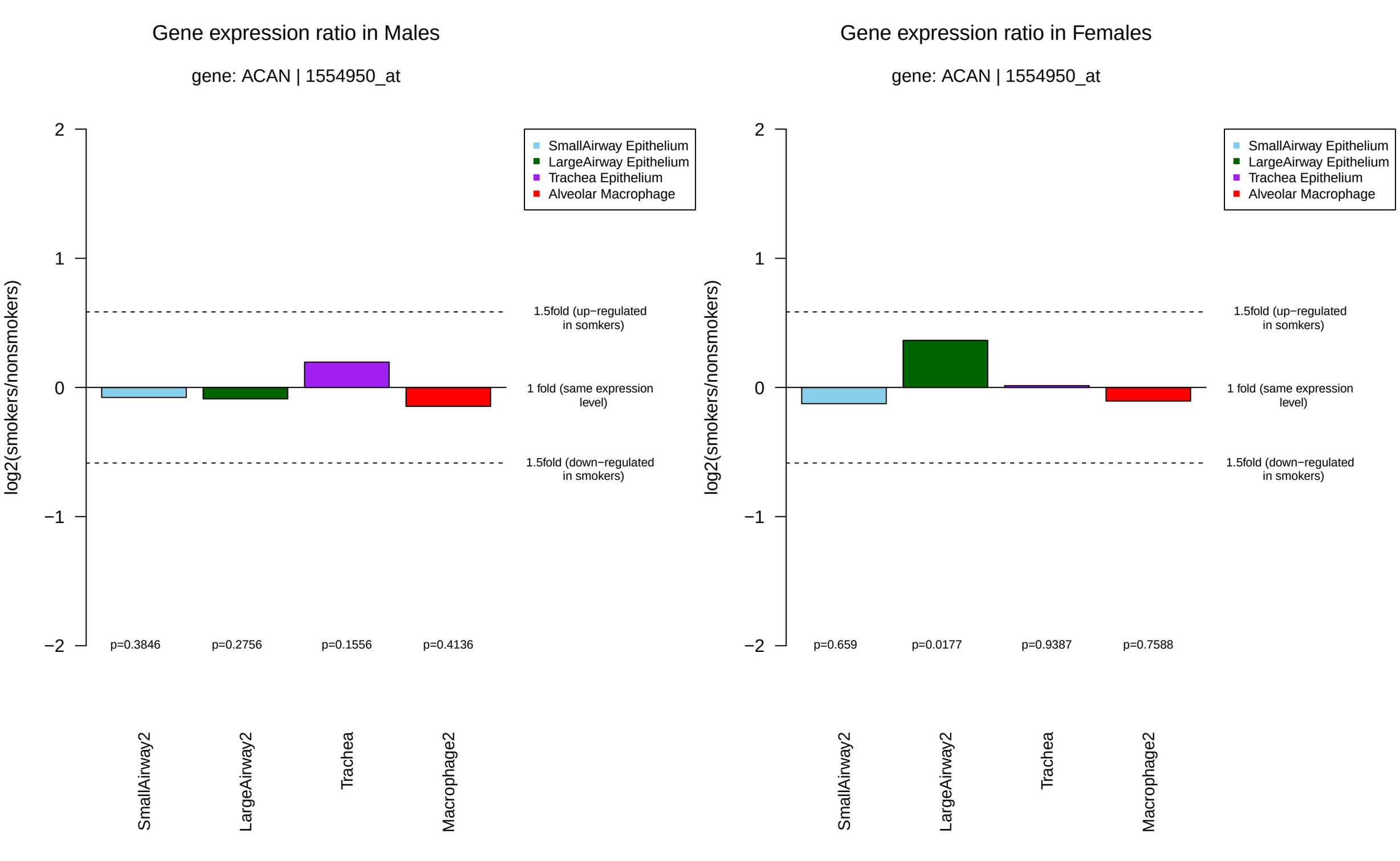
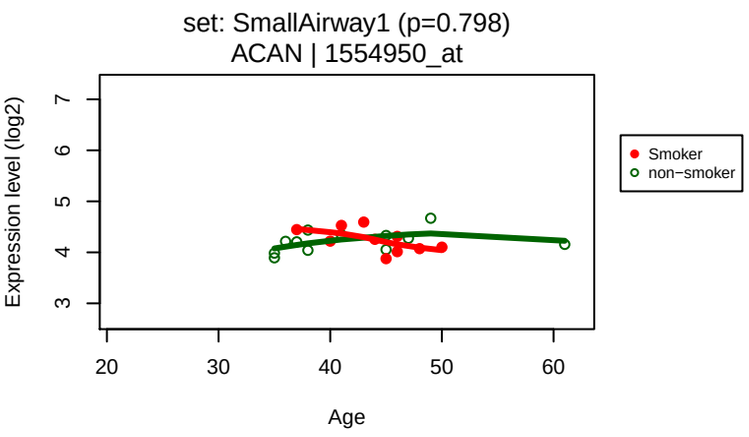
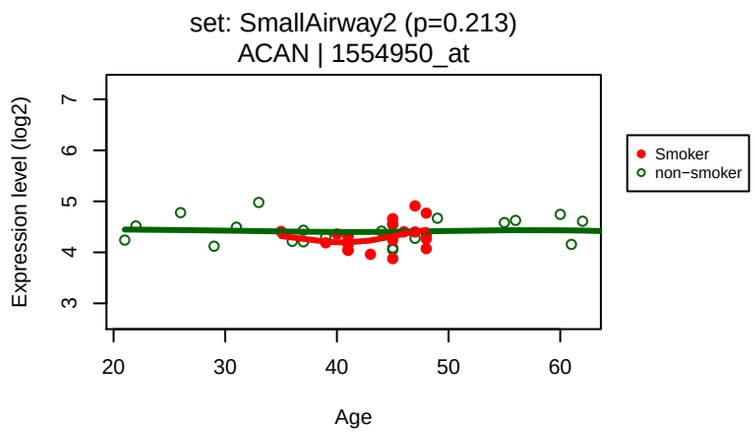
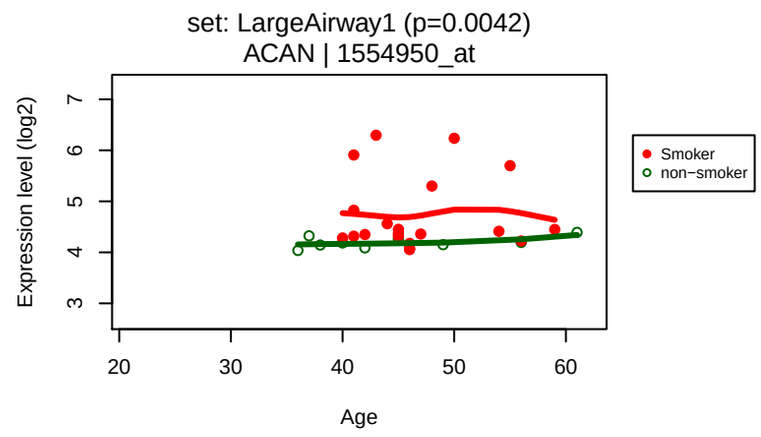

Gene expression of smokers and nonsmokers
gene: ACAN | 1554950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.798
p=0.213
p=0.5195
p=0.0042
p=0.3053
p=0.2288
p=0.8401
p=0.396
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

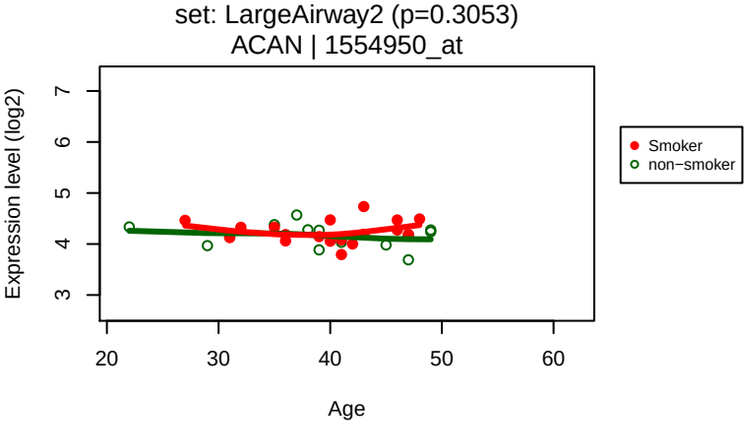

Gene expression ratio (smokers/nonsmokers)
gene: ACAN | 1554950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.798
p=0.213
p=0.5195
p=0.0042
p=0.3053
p=0.2288
p=0.8401
p=0.396
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

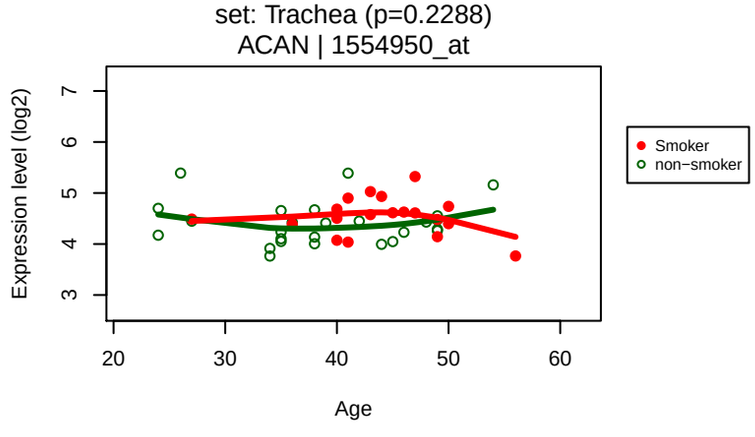

Gene expression ratio in Males
gene: ACAN | 1554950_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3846
p=0.2756
p=0.1556
p=0.4136
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACAN | 1554950_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.659
p=0.0177
p=0.9387
p=0.7588
SmallAirway2
LargeAirway2
Trachea
Macrophage2

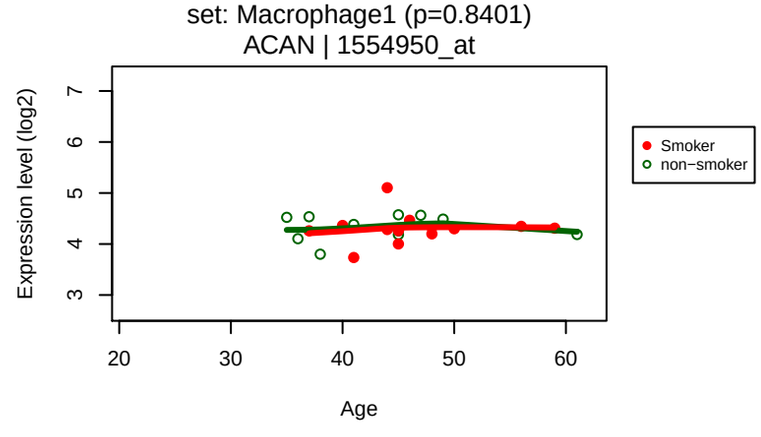

set: SmallAirway1 (p=0.798)
ACAN | 1554950_at
Expression level (log2)
Age
Smoker
non−smoker

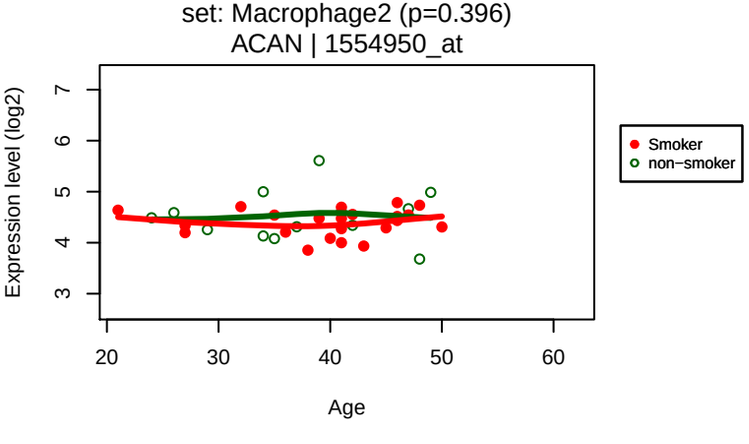

set: SmallAirway2 (p=0.213)
ACAN | 1554950_at
Expression level (log2)
Age
Smoker
non−smoker

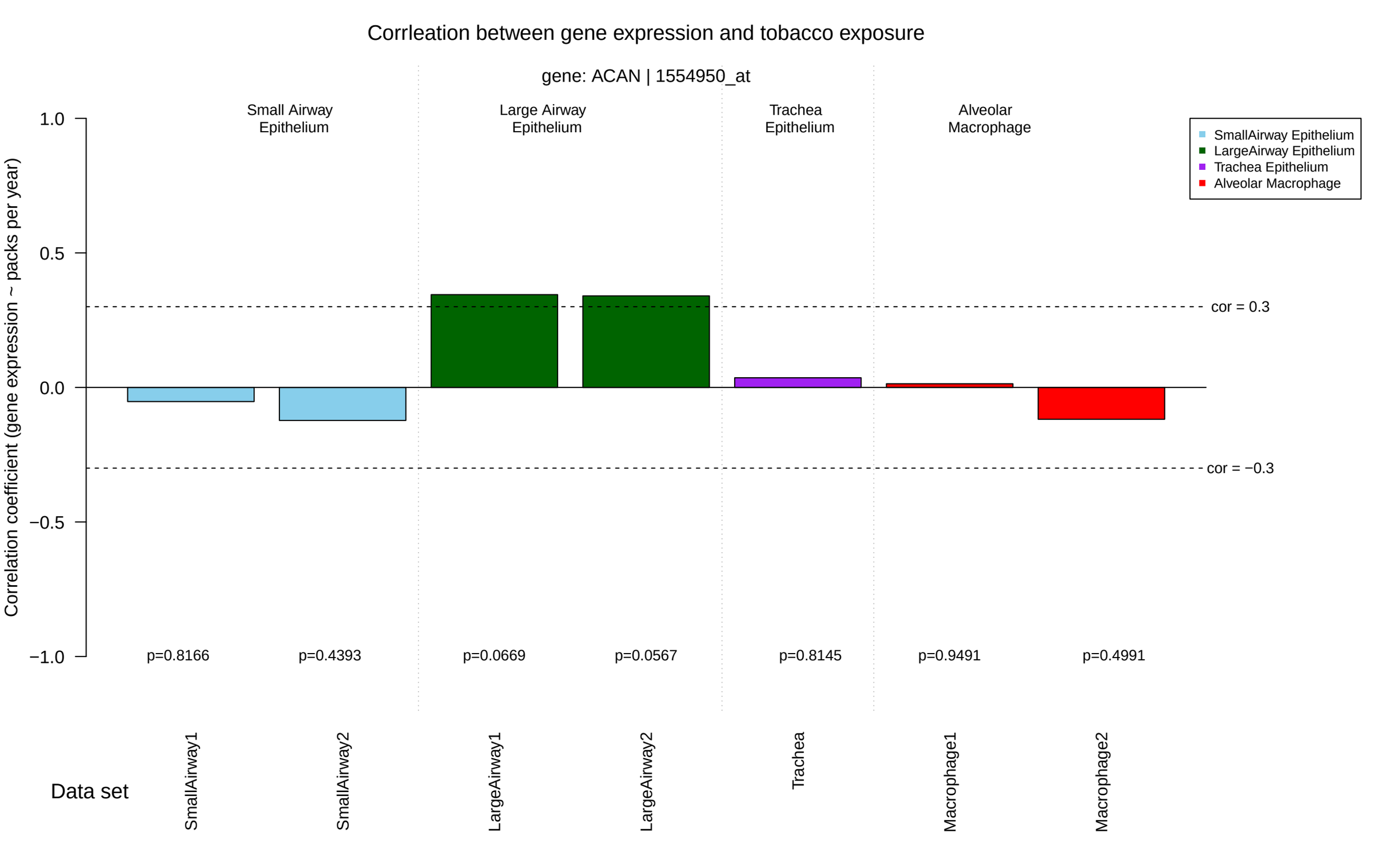

set: LargeAirway1 (p=0.0042)
ACAN | 1554950_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3053)
ACAN | 1554950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2288)
ACAN | 1554950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8401)
ACAN | 1554950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.396)
ACAN | 1554950_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACAN | 1554950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8166
p=0.4393
p=0.0669
p=0.0567
p=0.8145
p=0.9491
p=0.4991
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2